growing demands on compute infrastructure.

In 2011, Facebook shared its designs with the public and—along with Intel, Rackspace, Goldman Sachs and Andy Bechtolsheim—launched the Open Compute Project, incorporating the Open Compute Project Foundation. The five members hoped to create a movement in the hardware space that would bring about the same kind of creativity and collaboration we see in open source software. And that's exactly what's happening.

The Open Compute community has multiple active open hardware streams. There are multiple designs available for storage, networking, servers, racks and general guidelines for designing a data centre.

RISC-V is an open Instruction Set Architecture (ISA) and is delivered as a parameterised core generator. It has a small general-purpose base and multiple optional extensions, including reserved opcodes for unique SoC instructions.

Its goals are:

- To become the industry-standard ISA for all computing devices.
- To expand specifications for SoCs, including I/O and accelerators.
- As most of the cost in chip design is the cost of the software, this architecture will try to ensure that this software can be reused across many chip designs.
- Encourage both open source and proprietary implementations of the RISC-V ISA specifications.

RISC-V has been adopted by IIT Madras (and others too), and RISC-V chips are shipping commercially.

To summarise, enterprise class IT has evolved by adopting open source products and processes, resulting in more agility and helping organisations to stay ahead in the market with speedy innovations. It is no surprise that reducing operational IT expenditures, while simultaneously increasing the level of security and software capabilities, is a top priority for most enterprises. Open source addresses these concerns. Since enterprise open source comes with its own advantages and risks, one has to choose the right set of open source solutions to suit specific needs. Enterprises need to consider the advantages of open source and how it can help them improve their operations. END

References

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